

Lecture 12

Introduction to Random Variables and Mathematical Expectation

1 Random Variables

A random variable is a variable that takes on numerical values as a result of a random experiment or measurement, and associates a probability with each possible outcome.

The differences between variable and random variable are -

- Random variable always takes numerical values
- There is a probability associated with each possible values

Random variables are generally denoted by uppercase letters, such as X, Y, Z etc.

Example:

Consider the experiment of flipping two fair coins.

The possible outcomes are: $S = \{HH, HT, TH, TT\}$

Let X = The number of heads obtained in a trial

So, $X = \{0, 1, 2\}$

And the probabilities associated with each value of X can be represented by the following table:

$X = x$	0	1	2
$P(X = x)$	1/4	2/4	1/4

2 Types of Random Variables:

1. Discrete Random Variables: A random variable defined over a discrete sample space.

- X = Number of correct answers in a MCQ test with 100 marks = 0, 1, 2, ..., 100
- X = Number of cars passing a toll booth in a day = 0, 1, 2, ..., ∞
- X = Number of balls required to take the first wicket = 1, 2, 3, ..., ∞
- X = The number of phone calls received per day at a call centre = 1, 2, ... ∞

2. Continuous Random Variable: A random variable defined over a continuous sample space. Continuous random variables can represent any value within a specified range or interval and can take on an infinite number of possible values. Example:

- X = Weight of a person. $0 < X < \infty$
- X = Monthly Profit. $-\infty < X < \infty$
- X = Temperature recorded by the meteorological office . $0 < X < \infty$

3 Probability Functions of Random Variables:

Probability Mass Function (PMF)

The probability distribution function of a discrete random variable X is called a pmf and is denoted by $P(x)$.

PMF has the following properties:

1. $0 \leq P(x) \leq 1$
2. $\sum_{i=1}^n P(x_i) = 1$ for all x_i

Example 1:

Let X be a random variable with probability function defined as follows:

x	2	4	6	8
$P(x)$	$2/10$	$1/10$	$4/10$	$3/10$

1. $P(x = 4) = \frac{1}{10}$
2. $P(x > 4) = P(X = 6) + P(X = 8) = \frac{4}{10} + \frac{3}{10} = \frac{7}{10}$
3. $P(x \leq 6) = P(X = 2) + P(X = 4) + P(X = 6) = \frac{2}{10} + \frac{1}{10} + \frac{4}{10} = \frac{7}{10}$
4. $P(2 < x \leq 8) = P(X = 4) + P(X = 6) + P(X = 8) = \frac{1}{10} + \frac{4}{10} + \frac{3}{10} = \frac{8}{10}$
5. $P(x = 3) = 0$

Example 2:

A random variable X has the following probability distribution:

x	0	1	2	3	4	5	6	7
$P(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Determine i) k ii) $P(X < 3)$ iii) $P(X > 6)$ iv) $P(0 < X < 3)$

Example 3:

A random variable X has the following probability distribution:

x	0	1	2	3	4	5	6	7	8
$P(x)$	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

i) Determine the value of a ii) $P(X < 3)$ iii) $P(X \geq 3)$ iv) $P(0 < X < 5)$

Probability Density Function (PDF)

The probability distribution function of a continuous random variable X is called a pdf and is denoted by $f(x)$.

And a valid PDF $f(X)$ has the following properties:

1. $f(x) \geq 0$
2. $\int_{-\infty}^{\infty} f(x)dx = 1$
3. $P[a \leq x \leq b] = \int_a^b f(x)dx$

Example 4:

The probability density function of a random variable X is defined as

$$f(x) = \begin{cases} x; & 0 \leq x < 1 \\ 2 - x; & 1 \leq x < 2 \\ 0; & x \geq 2 \end{cases}$$

Find $P[0.5 \leq x \leq 1.5]$

Solution:

$$\begin{aligned} P(0.5 \leq x \leq 1.5) &= \int_{0.5}^{1.5} f(x)dx \\ &= \int_{0.5}^1 xdx + \int_1^{1.5} (2 - x)dx \\ &= \left[\frac{x^2}{2} \right]_{0.5}^1 + \left[2x - \frac{x^2}{2} \right]_1^{1.5} \\ &= \frac{3}{4} \end{aligned}$$

Example 5:

Suppose that X is a continuous random variable whose probability density function is given by

$$f(x) = \begin{cases} C(4x - 2x^2); & 0 < x < 2 \\ 0; & \text{otherwise} \end{cases}$$

- a) What is the value of C ?
- b) Find $P[x > 1]$

Example 6:

The probability density function of a random variable X given below:

$$f(x) = \begin{cases} kx; & 0 < x < 4 \\ 0; & \text{otherwise} \end{cases}$$

- a) What is the value of k ?
- b) Find $P[1 < x < 2]$ and $P[x > 2]$

4 Mathematical Expectation of Random Variables

Let X be a discrete random variable with pmf $p(x)$, or a continuous random variable with $f(x)$, the mathematical expectation of x is defined by

$$\mu = E[X] = \begin{cases} \sum x \cdot p(x); & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x \cdot f(x) dx; & \text{if } X \text{ is continuous} \end{cases}$$

Mathematical expectation is also known as population mean or expected value.

Example 7:

Let X be a random variable with probability function defined as follows:

x	2	4	6	8
$P(x)$	2/10	1/10	4/10	3/10

What is the expected value of X ?

Solution:

$$\begin{aligned} E(x) &= \sum x \cdot P(x) \\ &= (2 \times 2/10) + (4 \times 1/10) + (6 \times 4/10) + (8 \times 3/10) \\ &= 5.6 \end{aligned}$$

Example 8:

If the random variable X is the top face of a tossed, fair, six-sided die, what is the expected value of X ?

Solution:

$$X = \{1, 2, 3, 4, 5, 6\}$$

$$P(x) = 1/6; \quad \text{for } x = 1, 2, 3, 4, 5, 6$$

x	1	2	3	4	5	6
$P(x)$	1/6	1/6	1/6	1/6	1/6	1/6

$$\begin{aligned} E(x) &= \sum x \cdot P(x) \\ &= (1 \times 1/6) + (2 \times 1/6) + (3 \times 1/6) + (4 \times 1/6) + (5 \times 1/6) + (6 \times 1/6) \\ &= 3.5 \end{aligned}$$

Example 9:

Imagine a game in which, on any play, a player has a 20% chance of winning Tk. 30 and an 80% chance of losing Tk. 10. What is the expected gain/loss of the player in the long run?

Solution: Let X = the gain on a play

x	30	-10
$P(x)$	0.2	0.8

$$\begin{aligned} E(x) &= \sum x \cdot P(x) \\ &= (30 \times 0.2) + (-10 \times 0.8) \\ &= -2 \end{aligned}$$

Example 10:

Suppose X is a continuous random variable with probability density function

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

What is the expected value of X ?

Solution:

$$\begin{aligned} E(x) &= \int_0^1 x \cdot f(x) dx \\ &= \int_0^1 x \cdot 2x dx \\ &= \int_0^1 2x^2 dx \\ &= \left[\frac{2x^3}{3} \right]_0^1 \\ &= 2/3 \end{aligned}$$

Example 11:

The probability density function of a random variable X is defined as

$$f(x) = \begin{cases} x; & 0 \leq x < 1 \\ 2 - x; & 1 \leq x < 2 \\ 0; & x \geq 2 \end{cases}$$

What is the expected value of X ?

Solution:

$$\begin{aligned} E(x) &= \int_{-\infty}^{\infty} x \cdot f(x) dx \\ &= \int_0^2 x \cdot f(x) dx \\ &= \int_0^1 x \cdot f(x) dx + \int_1^2 x \cdot f(x) dx \\ &= \int_0^1 x^2 dx + \int_1^2 x \cdot (2 - x) dx \\ &= \int_0^1 x^2 dx + \int_1^2 (2x - x^2) dx \\ &= \left[\frac{x^3}{3} \right]_0^1 + \left[x^2 - \frac{x^3}{3} \right]_1^2 \\ &= 1 \end{aligned}$$

5 Properties of Mathematical Expectation:

Let X be a discrete random variable with probability mass function $P(x)$, or a continuous random variable with probability density function $f(x)$. Let $g(x)$ be a function of the random variable X . Then, the mathematical expectation of the function $g(x)$ is defined by

$$E[g(x)] = \begin{cases} \sum g(x) \cdot P(x); & \text{if } X \text{ is discrete} \\ \int g(x) \cdot f(x)dx; & \text{if } X \text{ is continuous} \end{cases}$$

For example, for a random variable X with probability function $f(x)$, the expected value of X^2 is

$$E[X^2] = \begin{cases} \sum x^2 \cdot P(x); & \text{if } X \text{ is discrete} \\ \int x^2 \cdot f(x)dx; & \text{if } X \text{ is continuous} \end{cases}$$

Let X and Y be two independent random variables, and let c be a constant. Then, the following properties are true:

- $E[c] = c$
- $E[cX] = cE[X]$
- $E[X + c] = E[X] + c$
- $E[X + Y] = E[X] + E[Y]$
- $E[X - Y] = E[X] - E[Y]$
- $E[XY] = E[X] \cdot E[Y]$

Example 12:

Let X denote a random variable that takes the values -1, 0 and 1 with respective probabilities

$$P(X = -1) = .2 \quad P(X = 0) = .5 \quad P(X = 1) = .3$$

Compute $E[X^2]$.

Solution:

$$\begin{aligned} E[X^2] &= \sum x^2 \cdot P(x) \\ &= (-1)^2 \cdot (0.2) + 0 \cdot (0.5) + 1 \cdot (0.3) \\ &= 0.2 + 0 + 0.3 \\ &= 0.5 \end{aligned}$$

Example 13: Let $Y = X + 2$ where X is the random variable from Example 12. Find the probability distribution of Y .

For $X = -1, 0, 1$; $Y = 1, 2, 3$ respectively

The probability distribution of Y :

y	1	2	3
$P(y)$	0.2	0.5	0.3

6 Variance of Random Variables

Variance:

$$\sigma^2 = \text{Var}(X) = E(X^2) - [E(X)]^2 = E(X^2) - \mu^2$$

Standard Deviation:

$$\sigma = SD(X) = \sqrt{\text{Var}(X)}$$

Example 13:

You want to open a new Café. After your market research, you found that 20% of similar cafés make a monthly loss of Tk. 50,000, 30% of them make no profit or loss, 40% make a profit of Tk. 50,000, and 10% of them make a profit of Tk. 150,000.

- What is your expected profit if you decide to open a new Café?
- What is the standard deviation in your profit amount?
- If your fixed cost increases by Tk. 10,000, what will be your new expected profit?

Solution: Let X = profit amount

$X = x$	-50,000	0	50,000	150,000
$P(x)$	0.2	0.3	0.4	0.1

a) $E(X) = (-50,000 \times 0.2) + (0 \times 0.3) + (50,000 \times 0.4) + (150,000 \times 0.1) = 25000$

b) $\sigma^2 = \text{Var}(X) = E(X^2) - [E(X)]^2$

$$E(X^2) = (-50,000^2 \times 0.2) + (0^2 \times 0.3) + (50,000^2 \times 0.4) + (150,000^2 \times 0.1) \\ = 3750000000$$

$$\therefore \text{Var}(X) = \sigma^2 = 3750000000 - (25000)^2 = 3125000000$$

$$\therefore SD(X) = \sigma = \sqrt{3125000000} = 55901.69944$$

c) $E(X - 10000) = E(X) - E(10000) = E(X) - 10000 = 25000 - 10000 = 15000$

7 Properties of Variance

Let X and Y be two independent random variables and c be a constant. Then, the following properties are true:

- $\text{Var}(c) = 0$

- $\text{Var}(cX) = c^2 \text{Var}(x)$

- $\text{Var}(X + c) = \text{Var}(x)$

- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

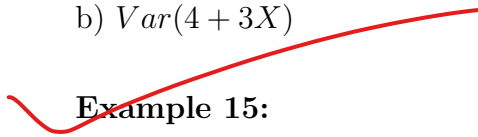
- $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$

Example 14:

If $E(X) = 1$ and $Var(X) = 5$ find

a) $E[(2 + X)^2]$

b) $Var(4 + 3X)$



Example 15:

A coin is biased so that the head is 3 times as likely to occur as tail. If the coin is tossed twice, find the probability distribution of number of tails.